

Madhur Rishi Sikarwar

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Education

R V College of Engineering

B.E. in Information Science and Engineering | CGPA: 9.33 / 10.0

Bengaluru, Karnataka

2024 – 2028

Technical Skills

Languages: C, C++, Java, Python, JavaScript, SQL, Solidity

Web & Backend: HTML, CSS, Flask, REST APIs, Full Stack Development

ML & AI: NLP, FinBERT, HuggingFace Transformers, PyTorch, Scikit-learn, Deep Learning

Data: NumPy, Pandas, Matplotlib, Google Earth Engine

Blockchain: Solidity, Smart Contracts, Ethereum, Web3.js, Truffle, Ganache

Tools: Git, GitHub, Linux, Google Colab, OpenSSL, ESP32, ThingSpeak

Core CS: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks

Projects

Algal Bloom Detection System 📶 IoT | 🛰️ Satellite

IoT & Remote Sensing

Python, Flask, ESP32, ThingSpeak, Google Earth Engine, Sentinel-2, NDCI, Chart.js

- Built a real-time monitoring dashboard ingesting live sensor data from a custom **ESP32 microcontroller** tracking 4 water quality parameters (pH, turbidity, TDS, dissolved oxygen) via ThingSpeak REST API.
- Designed a **weighted bloom-probability scoring algorithm** evaluating all parameters against configurable thresholds and triggering automated alerts on high-risk conditions.
- Architected a parallel cloud-based geospatial pipeline on **Google Earth Engine** processing **Sentinel-2 multispectral satellite imagery** to compute **NDCI** and produce georeferenced bloom-risk heat maps across multiple Indian water bodies.

Blockchain-Based Student Election System 📶 GitHub

Decentralized Application

Solidity, Ethereum Sepolia, Web3.js, MetaMask, Truffle, Ganache, JavaScript

- Deployed a fully decentralized voting platform on the **Ethereum Sepolia testnet**, eliminating central authority dependency and single points of failure.
- Authored **Solidity smart contracts** enforcing MetaMask wallet authentication, one-vote-per-address validation, anonymous ballot submission, and tamper-proof on-chain result tallying.
- Validated contract correctness and gas optimisation across all election lifecycle functions using Truffle and Ganache prior to testnet deployment.

Stock Market Sentiment Analyzer 📶 GitHub

NLP & Machine Learning

Python, FinBERT, HuggingFace Transformers, PyTorch, Pandas, Matplotlib, Scikit-learn

- Developed an end-to-end **NLP pipeline** using the domain-adapted **FinBERT transformer** for 3-class sentiment classification (bullish / bearish / neutral) on financial news headlines and social media posts.
- Engineered a preprocessing pipeline handling text cleaning, tokenisation, and batch inference via HuggingFace Transformers, significantly reducing manual labelling overhead.
- Aggregated per-source daily sentiment scores into composite market signals and rendered multi-series time-series trend visualisations correlated against asset price movements.

File Integrity Checker 📶 GitHub

Systems Programming & Cryptography

C, OpenSSL, SHA-256, SHA-512, MD5, POSIX File I/O, Linux

- Built a **command-line C utility** implementing MD5, SHA-256, and SHA-512 via OpenSSL with configurable hash-strength selection based on security requirements.
- Engineered recursive directory traversal using POSIX file I/O to serialise cryptographic baselines to a persistent manifest and detect unauthorised additions, deletions, or byte-level modifications.
- Produced structured timestamped change-log reports for forensic auditing, compliance verification, and system integrity monitoring.